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Analyzing cybersecurity gaps in medical device procurement using NLP and vulnerability databases

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Abstract

Medical devices are increasingly connected to public networks, offering benefits such as remote diagnosis but also introducing new cybersecurity risks. One pathway to strengthening protection at the design stage is to ensure that procurement requirements explicitly incorporate cybersecurity terms and controls. This study examines how such considerations are represented in medical procurement within Indian public healthcare sector. We compile and standardize 760 e-procurement documents (2014–2024) from major public hospitals across the country and analyze them using natural language processing techniques, including Term Frequency-Inverse Document Frequency (TF-IDF), Non-negative Matrix Factorization (NMF), and regular expressions, to identify security-related clauses. In parallel, 123 CISA advisories (2018–2024) highlighting vulnerabilities in medical systems are mapped to the MITRE Common Weakness Enumeration (CWE) framework. Bridging these two datasets (tender documents and CISA advisories) reveals significant gaps: cybersecurity specifications appear in only 15% (114/760) of tenders, and high-risk CWEs such as CWE-287 (Authentication Bypass) are largely unaddressed. Operational terms (e.g., “warranty”: 82%) dominate over technical controls (e.g., “encryption”: <5%). The findings indicate limited integration of threat intelligence into procurement design and suggest stronger policy measures for enhancing vendor accountability through security-aligned tender specifications.

Article Highlights

1. Text analysis of Indian medical e-procurement documents to assess coverage of high-risk cyber threats.
2. Results show focus on device maintenance, safety, and warranty, rarely mentioning known cyber risks.
3. Findings suggest e-procurement clauses could help introduce clear, enforceable cybersecurity requirements.

Keywords Text analysis, TF-IDF, Medical device vulnerabilities, Cybersecurity advisories, Tender analysis, Indian Medical e-procurements



1 Introduction

With the digitization of healthcare systems, the cybersecurity of medical devices has become a topic of paramount importance, critical not only for patient safety but also for sustaining trust in the healthcare ecosystem [15]. In response, substantial regulatory efforts have been made by national agencies through statutory requirements, guidelines, and standards, such as the EU Medical Devices Regulation (MDR) [42] and the 2022 amendments to the US Federal Food, Drug, and Cosmetic Act [49]. Similarly, broader data protection frameworks like the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA) [10] aim to establish robust privacy and data protection mandates. However, the responsibility for mitigating cybersecurity threats is a shared burden, distributed among regulators, device manufacturers, and healthcare providers.

One critical yet under-explored mechanism for enforcing this shared responsibility is the e-procurement process [34] 2007. Tender documents serve as a binding agreement between manufacturers and healthcare providers, specifying device requirements and evaluation criteria. This makes them a powerful, practical tool to mandate compliance with cybersecurity standards [40, 45]. The governance rationale for incorporating cybersecurity clauses in procurement documents is grounded in the concept that quality, safety, and security are coequal non-functional requirements in medical device tenders [24]. As medical devices become increasingly connected, cybersecurity is a critical component of patient safety and data protection. Procurement clauses can and should mandate cybersecurity measures, such as adherence to recognized standards (e.g., IEC 62304, ISO/IEC 80001), requirements for secure software development practices, and post-market vulnerability management. By explicitly including these clauses, healthcare institutions can enforce accountability from manufacturers and ensure that devices meet minimum security standards, thereby reducing the risk of cyber incidents that could compromise patient care.

Set against this backdrop, this paper presents a novel, data-driven analysis of e-procurement documents from Indian medical institutions (2014–2024) to evaluate the integration of cybersecurity into the procurement process. Specifically, we address the following research questions:

RQ1 Are remedial actions, either technical or regulatory, addressing frequent threats, vulnerabilities, and attacker tactics, explicitly included in e-procurements of Indian medical devices?

RQ2 To what extent are cybersecurity requirements explicitly mentioned in e-procurement documents? Are security mechanisms addressing recently reported vulnerabilities included in these documents?

To answer these questions, we conduct a dual-pronged analysis. We first analyse 123 Cybersecurity and Infrastructure Security Agency (CISA, 12 advisories (2018–2024) to establish a baseline of known vulnerabilities and threats. We then examine a corpus of e-procurement documents from leading Indian hospitals. By employing natural language processing techniques, including regular expressions and Term Frequency-Inverse Document Frequency (TF-IDF) [8], we bridge these two corpora to quantify the alignment between known threats and procurement mandates. Our findings reveal

a significant policy-practice gap, while regulatory compliance and device safety are emphasized, specific cybersecurity considerations addressing recent CISA-highlighted vulnerabilities remain conspicuously underrepresented. In line with open-science principles, all databases and scripts are made publicly available to ensure transparency and reproducibility.

1.1 Related work

Recent research has extensively documented significant cybersecurity vulnerabilities across various medical devices, including implantable devices [22], infusion pumps [26, 29], diagnostic systems like CT scanners [44], and life-support equipment such as ventilators and defibrillators [29]. The theoretical risk of these vulnerabilities has been validated by real-world incidents, including ransomware attacks like Wannacry [4] and NotPetya [19], major data breaches [37], and large-scale recalls of devices like the St. Jude pacemaker [17] and Medtronic MiniMed insulin pumps [25]. In response, regulatory agencies and standards bodies have developed frameworks to govern device security, including the U.S. FD&C Act, the EU MDR [42], and standards like IEC 62,304, complemented by data protection laws such as HIPAA and GDPR. While regulations and post-market surveillance define the “what” and detect failures, the critical link on how to ensure security by design has been overlooked. Research in other critical sectors, such as public infrastructure [34, 45] has demonstrated that procurement specifications are a powerful governance tool. Within the medical domain, this potential remains nascent. Only a few studies have begun to explore this intersection. [6] conducted a high-level analysis of meta-information from Open Contracting Data Standards across 36 countries to flag potentially insecure equipment. In a complementary approach, [43] evaluated the inclusion of cybersecurity content in standardized FDA product summaries.

Our study diverges from this emerging literature in two critical ways, offering a more granular and contextually grounded analysis. First, while Bracciale et al. relied on meta-data, we analyze the full text of over 760 procurement documents, sourced directly from Indian hospital websites. This provides a deeper, more nuanced understanding of the actual security requirements mandated in contracts, moving beyond high-level flagging. Second, unlike Stern et al. focus on standardized FDA summaries, we analyze the inherently diverse and less-structured language of real-world tender documents. This captures the practical, on-the-ground priorities of healthcare providers, offering a richer context for understanding how cybersecurity is integrated or neglected in procurement practices. To our knowledge, this work presents the first comprehensive, data-driven analysis of cybersecurity content in Indian medical e-procurement, systematically bridging the gap between known vulnerabilities and procurement mandates to suggest policy measures improving the sector.

1.2 Overview of the paper

Rest of the paper is divided into five sections. Section 2 discusses the techniques of text mining and topic modeling of text analysis. Section 3 presents the comprehensive methodological framework, describing the study design, computational environment, and reproducibility measures that support the research findings. Section 4 focuses on the analysis of CISA advisories related to medical devices, highlighting prevalent threats,

vulnerabilities, and attacker tactics. Section 5 delves into the e-procurement documents, sourced through web scraping of Indian hospital websites, to examine their cybersecurity content. Section 6 provides a comparative analysis of both corpora, identifying key omissions and gaps, and offering recommendations for advancing current practices. At last, Sect. 7 concludes the paper by summarizing the key findings and proposing actionable directions for future work.

2 Background on text processing and topic modelling

Extracting crucial information in a large corpus of (un)structured documents is a strenuous but essential task, useful in several domains for example, bibliometrics [7], market intelligence [41], opinion mining [21], discovery of hidden insights [3], ontology [27]. Numerous tools and techniques for automatically acquiring knowledge from the text have been proposed by the natural language processing community using expert systems [28], machine learning [18], deep learning [33] and graph-based algorithms [50]. Popular models and their adaptations include Cyner [1] and Cybert [2]. However, they typically require extensive annotated datasets, which are unavailable in our context. Hence in this work, we utilize regular expressions as a computationally efficient alternative. Alongside, we employ TF-IDF (Term Frequency-Inverse Document Frequency) and Non-negative Matrix Factorization (NMF) to determine “cybersecurity topics” that can be inferred from the corpus.

Below, we briefly describe these two techniques:

A. Term Frequency-Inverse Document Frequency (TF-IDF) TF-IDF is a term weighting scheme commonly used to represent textual documents as vectors for classification, clustering, visualization, retrieval, etc. In this paper, we employ the technique to determine whether cybersecurity considerations are explicitly included in the procurement criteria and, if so, to what extent. Let $T = \{t_1, \dots, t_n\}$ be the set of all terms occurring in the document corpus under consideration. Then a document d_i is represented by an n -dimensional real-valued vector $X_i = (x_{i1}, \dots, x_{in})$ with one component for each possible term from T . The weight x_{ij} corresponding to term t_j in document d_i is a product of three parts: one which depends on the frequency of t_j in d_i , one which depends on t_j presence in the corpus as a whole d_j . TF-IDF weighting is defined by: $x_{ij} = TF_{ij} \cdot IDF_j \cdot \left(\sum_j (TF_{ij} \cdot IDF_j)^2 \right)^{-\frac{1}{2}}$

where TF_{ij}

is the term frequency (i.e., number of occurrences) of t_j in d_i , and IDF_j is the IDF of t_j , defined as $\log \left(\frac{N}{DF_j} \right)$, where N is the number of documents in the corpus and DF_j is the document frequency of t_j (i.e., the number of documents in which t_j occurs). The normalization part ensures that the vector has a Euclidean length of 1.

In this paper, we implement TF-IDF using the Sci-kit Python library and pandas (in Subsection 4.3 and 5.2).

B. Non-negative Matrix Factorization (NMF) We use NMF (Non-negative Matrix Factorization) to note the security themes in the vulnerability advisories. This is done by applying the numerical TF-IDF matrix as discussed in previous paragraph to extract latent topics from the data. The NMF algorithm decomposes the TF-IDF matrix into two matrices:

- *Document-Topic Matrix (W)* This matrix represents the relationship between each document and the topics, indicating the extent to which each document is associated with different themes. It provides a representation in which each document elucidates the themes identified by the NMF, and the values indicate the strength or weight of each topic within that document. It allows us to categorize documents based on their predominant themes.
- *Topic-Term Matrix (H)* This matrix represents the relationship between topics and terms, showing the most significant terms for each topic. The values in this matrix indicate the importance of each term for defining the themes, which helps understand the meaning behind each topic by listing the terms that contribute most significantly to it.

In this process, the NMF algorithm finds hidden patterns within the TF-IDF matrix by grouping similar terms and documents. Using the Document-Topic Matrix (W), we can identify which topics are present in each document and to what degree. NMF allows us to cluster documents containing similar theme profiles, providing insights into the categories of vulnerabilities or issues under discussion. Using the Topic-Term Matrix (H), we can identify the most representative terms for each theme. Using the most relevant terms, we can define and interpret the characteristics of each theme.

3 Methodology

Figure 1 shows methodological flow of our work. It employs a dual-stream analytical approach to investigate the cybersecurity gap between known vulnerabilities in medical devices and the requirements specified in public procurement documents. The methodology is structured into two parallel workflows: one for analyzing ICS security advisories and another for analyzing e-procurement tenders.

3.1 Computational environment & key libraries

All analyses were conducted in a Python 3.12.3 environment [36] on a 64-bit Linux system (Kernel 6.8.0–84.0-generic, GCC 13.3.0). Key computational libraries included *pandas* (v2.3.3) [31] for data manipulation, *scikit-learn* (v1.7.2) [35] for machine learning and natural language processing tasks, and *NLTK* (v3.9.2) [5] and *spaCy* (v3.8.7) [47] for linguistic processing. Web data collection relied on the *Requests* [38] and *BeautifulSoup* [39] libraries to retrieve and parse advisory and procurement documents. The standard Python *re* module was used to extract and normalize vulnerability identifiers (e.g., CVE, CWE). *NumPy* (v1.26.4) [46] supported numerical operations, while *pandas* [31] provided efficient tabular data handling. For visualization, *matplotlib* (v3.10.0) [23] and *seaborn* (v0.13.2) [48] were employed to create descriptive and comparative plots. Topic modeling and text weighting were performed using *scikit-learn*'s TF-IDF and non-negative matrix factorization utilities [35]. Topic coherence in NMF is calculated using the “Gensim library (version 4.4.0)” in Python, which implements the Coherence Model function. The “c_v metric” was employed to quantify the semantic interpretability of the NMF-derived topics. Finally, *ydata-profiling* (v4.17.0) [11] facilitated exploratory data analysis and quality assessment. These open-source tools were selected for their robustness, reproducibility, and established use in cybersecurity data workflows.

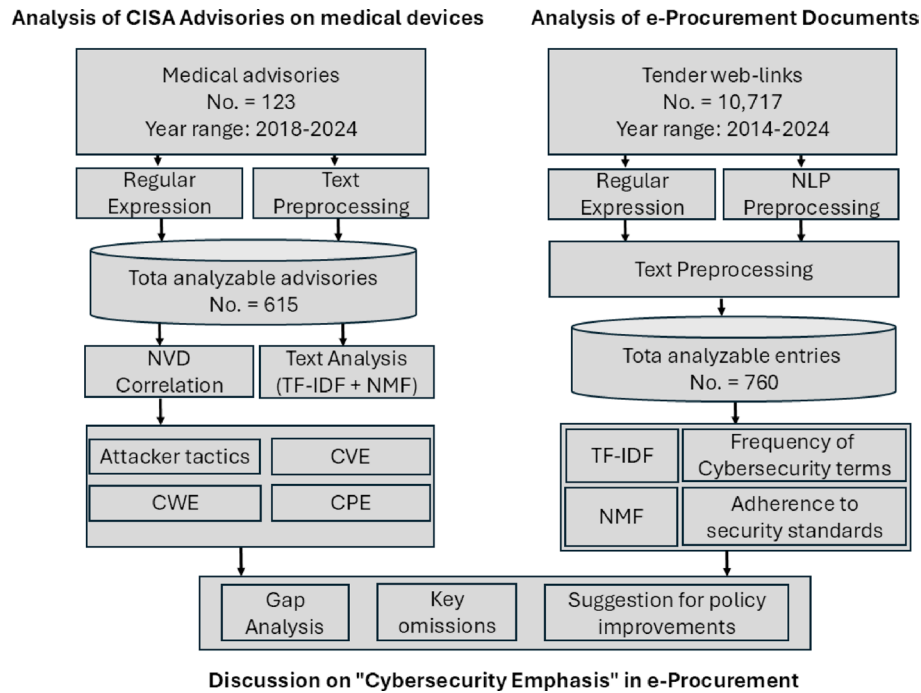


Fig. 1 Methodology followed in the paper showing how medical advisories and e-procurement documents were analysed to understand cybersecurity gaps in medical devices

3.2 Study design

The study sampling frame comprises two complementary data sources: Industrial Control Systems (ICS) medical advisories issued by the Cybersecurity and Infrastructure Security Agency [12] and procurement documents collected from the official e-tender portals of 17 government-run All India Institutes of Medical Sciences (AIIMS) across India. The choice of AIIMS was motivated by their national representation and the standardization of procurement practices across these institutions. However, we acknowledge that focusing solely on public tertiary care hospitals in India may introduce sampling bias, as procurement practices in private hospitals or at primary and secondary care levels may differ. Additionally, the Indian healthcare context may have unique regulatory and operational characteristics that limit immediate generalizability to other countries. Nevertheless, the methodological framework of comparing vulnerability databases with procurement documents is generalizable and can be applied in other contexts. The study's findings are representative of the Indian tertiary public healthcare sector, which is a significant segment of the Indian healthcare system.

3.3 Methodological considerations & reproducibility

In this paper, we used natural language processing methods of regex, TF-IDF, and NMF. We extracted cybersecurity terms such as of CVE/CWE ID and standards such as ISO/IEC by defining generic expressions (regex) to capture such information. For this purpose, we use a comprehensive dictionary on cybersecurity terms provided in [20]. While our approach to use regex seems technically naive, other popular machine learning based approaches such as Named Entity Recognition (NER), or Semantic Role Labeling (SRL), requires large amount of unambiguous and many times annotated data, while in our case the target patterns are well-defined, consistent, and relatively simple. Similarly,

we use TF-IDF, which is often considered a better choice than traditional techniques such as plain term frequency (TF) or raw count vectorization [9]. For topic modeling of both the corpora, we employed Non-Negative Matrix Factorization (NMF). Another popular technique is Latent Dirichlet Allocation (LDA). We choose NMF as unlike LDA's probabilistic generative approach, it is a statistical matrix factorization method that scales efficiently to large, technical corpora and produces more stable, interpretable topics [16]. Its non-negativity constraint ensures additive word contributions, reducing noise, and enhancing thematic clarity.

Our approach can be reproduced, from collecting unfiltered data to reproducing the results of the filtered data using the GitHub repository [32]. The dataset present in the repository can be used for other types of analysis, such as annual trend identification of ICS medical advisory as well as of E-procurement documents. The resources in the repository include the scraping code, the filtering approach, and reproducible analysis of data.

4 ICS advisory analysis

4.1 Data collection & preprocessing

Our collection includes 123 advisories issued between 2018 and 2024 (last accessed on 2024-11-03). Advisory web pages were retrieved using the *Requests* library and parsed with *BeautifulSoup* to extract key fields, including vendor information, equipment models, CVSS scores, and vulnerability descriptions. The unstructured data was initially organized into a *panda DataFrame* and then processed through a multi-stage cleaning pipeline. Standardized vulnerability identifiers, CVE (CVE-\d{4}-\d{4,7}), and CWE (CWE-\d+) were systematically extracted using regular expressions. Pandas was also used to split combined vendor and equipment names into separate rows to better organize the data. Using regex, we fixed inconsistencies in vendor names by identifying vendors with different versions of the same name in the advisory list and replacing such repeated names from 123 advisories with one standard name expanded the total advisory count to 128. Next, we combined different text columns 'URL', 'CVSS', 'Vendor', 'Equipment', 'Vulnerability', 'Risk Evaluated', 'Affected Product', 'Vulnerability Overview', 'Researcher', 'Mitigation' into one large text column, cleaned it by removing symbols and short words. The Equipment column sometimes had multiple items listed together, so we used the *re* library to split each entry in that column into separate pieces based on different separator (, | ; | \band\b | &) "exploding" the dataset to create individual rows for each unique device, yielding 203 entries. For thematic analysis, textual fields (e.g., Vulnerability Overview, Mitigation) were merged into a unified corpus, cleaned of extraneous symbols and stop words, and processed using two scikit-learn (version: 1.7.2) libraries:

- *TfidfVectorizer* to convert text into numbers based on word importance.
- NMF to discover common topics across all the data using the numerical representation from *TfidfVectorizer* to gain topic insight.

A final quality check removed records with critical missing values, producing a robust dataset of 615 unique equipment-vulnerability pairs suitable for in-depth analysis.

Table 1 Top CWEs identified in CISA advisories

CWE ID	CWE name	Count
CWE-287	Improper authentication	59
CWE-798	Use of hard-coded credentials	46
CWE-330	Use of insufficiently random values	41
CWE-20	Improper input validation	34
CWE-522	Insufficiently protected credentials	31
CWE-120	Buffer copy without size checking	29
CWE-79	Improper neutralization of input during web page generation	27
CWE-319	Clear text transmission of sensitive information	26
CWE-306	Missing authentication for critical function	21

Table 2 Top five vulnerable medical equipment identified in CISA advisories

Equipment	Vendor	Usage
Vue PACS	Philips	System for managing medical imaging data.
Agilia connect infusion system	B. Braun Melsungen AG	Medical infusion system for delivering medication.
Battery pack SP with Wi-Fi	Fresenius kabi	Operates stand-alone pumps in modern IT infrastructure.
Openclinic GA	Sourceforge	Open-source hospital management software.
SpaceCom	Philips	Satellite communication equipment for medical data.

4.2 Vulnerability enrichment & analysis

We analyze the advisories, using regular expressions for the identification and extraction of specific patterns within strings, making them particularly useful for parsing complex data formats.

Table 1 highlights the top 10 CWEs, offering insight into the most frequent weaknesses in medical devices. CWE-287 (Improper Authentication) emerged as the most prevalent issue, signaling persistent vulnerability in identity verification processes that could allow unauthorized access to critical medical equipment. Similarly, CWE-798 (Use of Hard-coded Credentials) ranked among the top vulnerabilities, indicating widespread reliance on hard-coded credentials as a significant security flaw. Attackers can easily exploit hard-coded credentials to gain control over devices. These vulnerabilities align with broader trends in cybersecurity, as both CWE-287 and CWE-798 are also prominent in the OWASP Top 10 list, which focuses on common web security flaws. Their frequent appearance in medical advisories underscores a consistent weakness in access control and credential management, making these systems highly susceptible to exploitation. Table 2 lists the top five most vulnerable medical devices, highlighting their vendors, usage, and the count of vulnerabilities identified. We identified that Vue PACS by Philips is the most frequently mentioned equipment with 19 vulnerabilities.

To note the severity of the analysed vulnerabilities, we enrich the dataset by cross-referencing CVE IDs with the National Vulnerability Database (NVD) to append meta-data on attack vectors and attack complexity. Figure 2 presents the results in form of a heatmap, how vulnerabilities are distributed across different attack vectors and severity levels. The rows represent attack vectors such as ADJACENT_NETWORK, LOCAL, NETWORK, and PHYSICAL, while the columns depict the severity levels: Critical, High, Medium, and Low. The heatmap reveals that most NETWORK vulnerabilities are deemed Critical (155 vulnerabilities) or High (85 vulnerabilities) severity levels. This

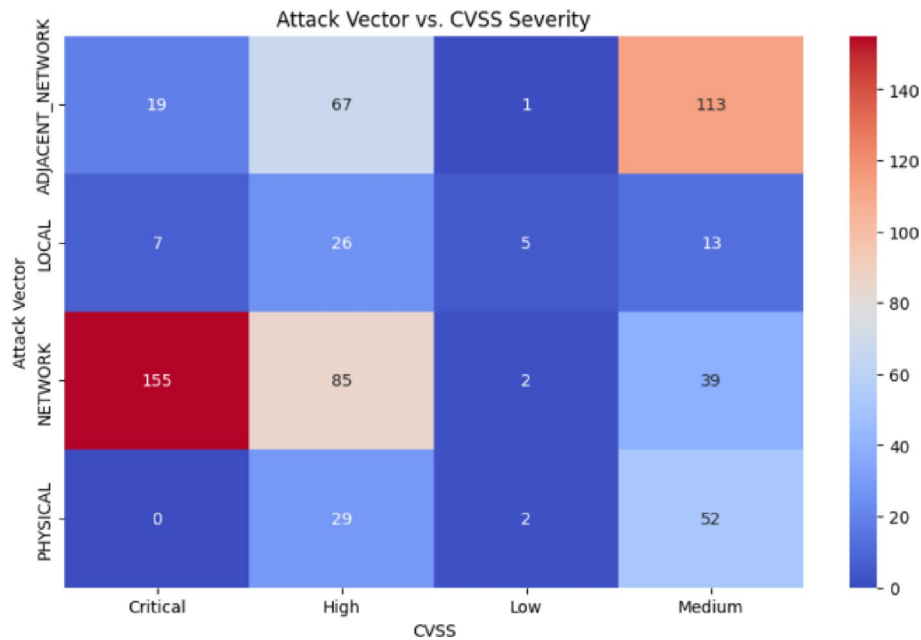


Fig. 2 Heatmap showing the distribution of vulnerabilities by attack vector and CVSS severity level

analysis indicates that NETWORK-based vulnerabilities are more common and tend to be more severe compared to other attack vectors.

Table 1 highlights the top 10 CWEs, offering insight into the most frequent weaknesses in medical devices. Table 2 lists the top five most vulnerable medical devices identified using known attack vectors, which define how the device can be attacked.

4.3 Topic modeling and analysis (TF-IDF and NMF)

The Non-Negative Matrix Factorization revealed five distinct thematic patterns within the ICS security advisories, each representing significant cybersecurity concerns in medical devices:

- *Topic 1: Social Engineering and Human Factor Vulnerabilities* - This theme highlights the prevalence of social engineering attacks, particularly email-based threats targeting ICS systems. The identified terms emphasize critical human factor vulnerabilities, underscoring the necessity for enhanced security awareness training and procedural controls in healthcare environments.
- *Topic 2: Remote Access and Network Security Gaps* - Remote access vulnerabilities emerge as a persistent concern, with specific focus on Virtual Private Network (VPN) implementations and network configuration weaknesses. This pattern indicates systemic security deficiencies in how remote management capabilities are deployed and maintained across medical and industrial control systems.
- *Topic 3 & Topic 5: Vendor-Specific Vulnerability Patterns* - Distinct vulnerability clusters are associated with major medical device manufacturers. Philips devices demonstrate issues primarily in device management systems, often addressed through specific security patches, while Medtronic equipment, particularly insulin pumps and patient monitors shows susceptibility to network-based and wireless exploitation vectors.

- *Topic 4: Software Integrity and Memory Management* - Buffer overflow vulnerabilities in TCP stack implementations, especially within the VxWorks operating system, represent significant software integrity risks. This theme highlights critical weaknesses in medical device communication protocols that require urgent attention in secure software development practices for industrial applications.

5 Medical e-procurement document analysis

5.1 Data collection & preprocessing

E-procurement (electronic procurement, sometimes also known as supplier exchange) is the business-to-business, business-to-consumer, or business-to-government purchase and sale of supplies, work, and services through the internet along with other information and networking systems, such as electronic data interchange and enterprise resource planning. Several authors, for example, [14] have highlighted the benefits of open e-procurement databases in promoting transparency, accountability, efficiency, and societal participation in public management. Consequently, governments globally have developed web portals with open procurement information, for example, the Open procurement platform, developed within the framework of the Digital Whistleblower (DIGIWHIST) project. This platform allows searching and analyzing public procurement processes for the 28 EU member countries, Norway, Iceland, Switzerland, Georgia, and EU institutions. Other country-specific government e-procurement platforms are available, for example in the USA.

In India, the Central Public Procurement Portal¹ (CPPP) is a standardized platform that facilitates the repositioning and free access to detailed procurement information relevant to government departments. While the information is exhaustive, in addition to the portal, we refer to various hospital e-procurement websites for more contextual information. In particular, we referred to the public procurement data of India's apex medical hospital, the "All India Institute of Medical Science"². The availability of detailed technical specification procurement documents is one of the reasons for geographically limiting this study to the Indian context. Python scripts with Selenium WebDriver were used to automate navigation and extract procurement metadata, resulting in a dataset of 10,717 unique URLs, including titles, document links, and publication dates, which were subsequently filtered using domain-specific keywords to isolate tenders pertaining to digital medical equipment, reducing the set to 2,531 documents. Non-textual files and documents shorter than three pages were excluded during quality assurance, yielding 1,300 candidate documents. Full text extraction from these PDFs was performed using a Python-based script with the pdfplumber library, chosen for its accuracy in extracting text, tables, and metadata from complex layouts.

The manual screening of the 1,300 candidate documents was conducted using a structured annotation protocol to ensure consistency. Prior to the review, all annotators participated in a training session using a shared codebook that operationalized the inclusion criteria with explicit definitions of 'digital medical equipment.' To validate the consistency of the screening process, a subset of documents was reviewed by multiple annotators, and discrepancies were resolved through consensus discussions led by the primary researcher. This iterative calibration ensured a unified understanding and application

¹ <https://eprocure.gov.in/eprocure/app>.

² <https://pmssy.mohfw.gov.in/index.php>.

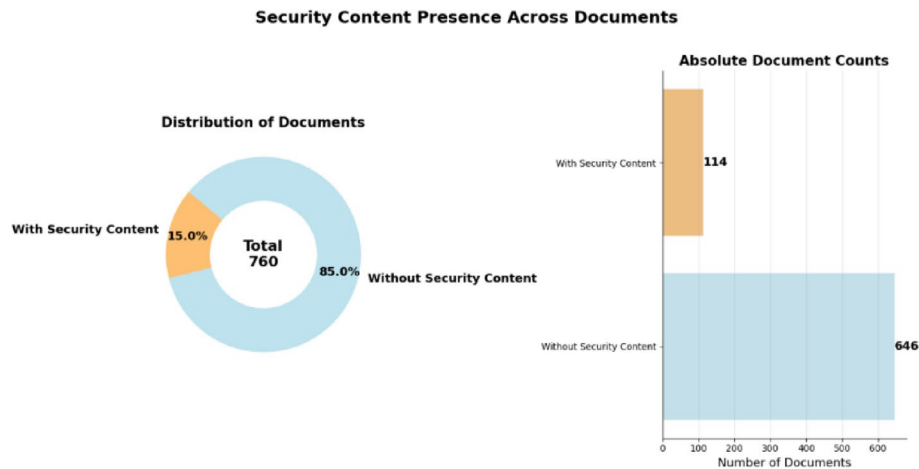


Fig. 3 Heatmap showing the distribution of vulnerabilities by attack vector and CVSS severity level

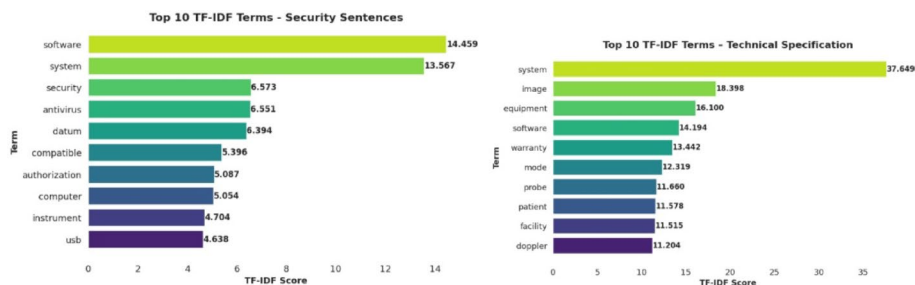


Fig. 4 Bar chart counting the frequent terms (y-axis) vs. their count (x-axis) across only technical specification and cybersecurity content

of the coding criteria throughout the screening process. The document refinement followed a transparent sequence: initial keyword filtering identified digitally-relevant procurements (2,531 documents), quality checks excluded non-textual and abbreviated documents (1,300 documents), and finally, manual verification confirmed device relevance and extracted technical specifications (760 documents).

5.2 Content analysis

We perform a preliminary analysis of the procurement documents using keyword search of the popular cyber-security themes terms from the OWASP glossary, combined with sentence boundary detection. Figure 3 reveals a key insight: only 15% of the documents reference cybersecurity terms, indicating limited focus on cybersecurity considerations.

To further explore significant terms, we used a term frequency-based analysis, filtering sentences containing cybersecurity-related terms. The resulting bar chart (Fig. 4 (left-hand)) highlights key themes such as *authorization*, *software*, *security*, and *system*, emphasizing system security, software protection, data integrity, and access control mechanisms. These terms underline essential cybersecurity considerations for safeguarding medical devices and their integration into hospital systems. In contrast, the same analysis of the entire corpus of technical specification (Fig. 4 (right hand)) reveals a different focus. Prominent terms like *system*, *image*, *inquiry*, *software*, *equipment*, and *warranty* reflect attention to equipment specifications, software requirements, and operational details. Additionally, terms like *probe*, *monitor*, and *Doppler* highlight the

specificity of medical instruments, while *warranty* and *supply* point to concerns about product guarantees and delivery expectations.

Figure 5 presents a two-dimensional semantic mapping of terms extracted from Indian medical e-procurement documents, across two dimensions: Function (vertical axis) and Specificity (horizontal axis). The horizontal dimension differentiates vague from precise terminology, with terms on the right (e.g., equipment, software, certificate) denoting clearly defined requirements and those on the left (e.g., security, mode, datum) indicating ambiguous or context-dependent, often lacking measurable or enforceable meaning within tender documents. The vertical axis represents Function, distinguishing between Technical Control terms and Governance/Process terms. Terms located in the upper half (e.g., order, duty, payment, warranty) relate to procurement governance, administrative procedures, and compliance obligations. In contrast, those in the lower half (e.g., antivirus, probe, computer, software) pertain to technical controls, operational components, or device-level specifications. The resulting four-quadrant distribution illustrates how procurement discourse privileges administrative precision over technical clarity. Terms such as payment, certificate, and warranty cluster in the Governance-Precise quadrant, reflecting well-defined financial and procedural requirements. In contrast, security-relevant terms like security, antivirus, and mode appear in the Technical-Vague quadrant, indicating that cybersecurity is mentioned but not operationalized. Even when technical elements such as system and equipment are articulated precisely, they rarely co-occur with security-oriented language. This imbalance highlights a linguistic and governance gap in the e-procurement process, where administrative obligations are formally codified but cybersecurity requirements remain under-specified and weakly enforceable. Additionally, employing bigrams (combinations of two consecutive words) based on the most frequent terms facilitates a better understanding of contextual meaning that single words alone fail to capture. Security sentences extracted from the

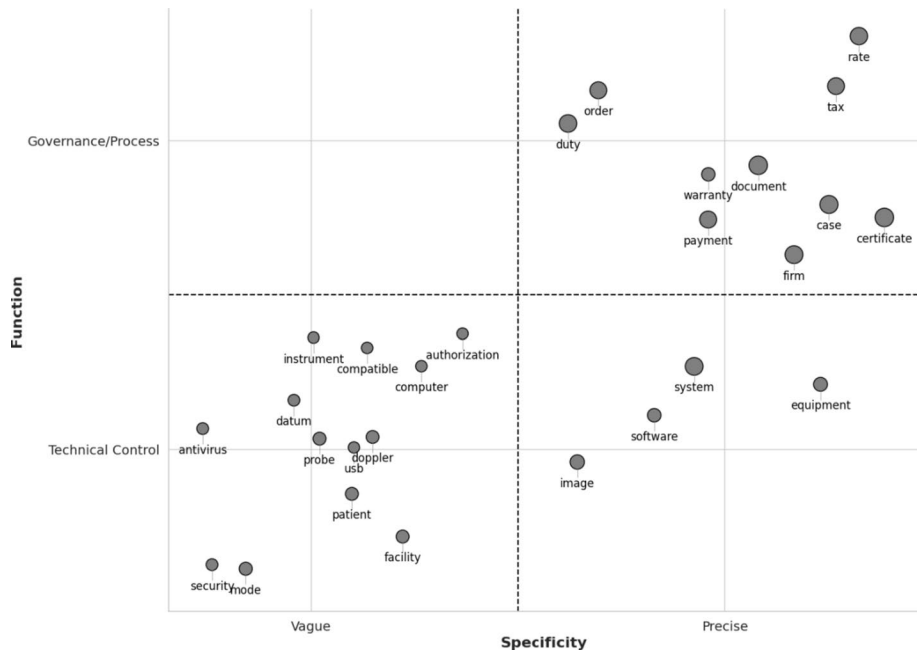


Fig. 5 Bar chart counting the frequent terms (y-axis) vs. their count (x-axis) across only technical specification and cybersecurity content

technical specification emphasizes on access control, antivirus software, authentication and authorization certificate, and encryption. Here, the first few terms related to security aspect, but terms such as authorization certificate are for asking the distributor to provide a certificate of authorization to sell the product from the manufacturer and is not related to cybersecurity

For noting the thematic structure emerging, we use Non-Negative Matrix Factorization (NMF) analysis of the technical specification corpus. Our NMF model achieved a topic coherence score of 0.7027 (c_v) on technical specification content, which reflects strong semantic alignment among the terms within each topic. The security-related subset of the procurement corpus yielded a topic coherence score of 0.4228, suggesting that cybersecurity-related terms form broader thematic clusters with less frequent co-occurrence patterns. The NMF analysis reveals that procurement language clusters around nine functional themes spanning a broad technological spectrum, from imaging and diagnostic systems (Topics 1, 2, 5, 6) to clinical instrumentation and monitoring platforms (Topics 4, 7, 9), as well as operational and administrative categories (Topics 3, 8). From a practical perspective, these themes highlight the strong *functional orientation* of procurement language. Specifications tend to emphasize performance, reliability, and maintenance, as evidenced by terms such as *warranty*, *maintenance*, and *preventive*, over security-centric attributes like *encryption*, *authentication*, or *access control*.

5.3 Adherence to cybersecurity standards

Healthcare, as a critical infrastructure sector, operates under extensive compliance regimes emphasizing quality, safety, and reliability. Within the analyzed procurement corpus, these priorities are clearly reflected in the distribution of cited standards.

Tables 4 and 5 Count of safety/security standards across the e-procurement document corpus.

Table 3 provides a list of ISO standards sorted based on their decreasing acronym frequency. In Table 4, the IEC standards are grouped by category to highlight the focus areas within the analyzed procurements. ISO 9001 and its variants (49 mentions) dominate, underscoring the primacy of quality management, while ISO 13,485 (11 mentions) and ISO 14,971 indicate strong attention to medical device quality assurance and risk management. In contrast, cybersecurity-oriented standards appear far less frequently. ISO/IEC 27,001 and ISO/IEC 27,002 are referenced only six times in total, representing

Table 3 Topics extracted from the technical specification corpus using NMF

Topic	Topic name	Keywords
1	Imaging and camera systems	video, image, camera, light, system, definition, output, high
2	Ultrasound and doppler systems	doppler, transducer, imaging, system, mhz, image, mode, probe
3	Equipment maintenance	equipment, site, maintenance, warranty, modification, preventive, hospital, basis
4	Patient monitoring systems	monitor, monitoring, pressure, ecg, alarm, patient, display, parameter
5	PCR and analysis systems	system, sample, pcr, test, temperature, plate, analysis, instrument
6	Microscopy systems	microscope, fluorescence, stage, camera, filter, objective, plan, light
7	Surgical navigation systems	system, navigation, probe, instrument, registration, surgery, software, image
8	Medical department equipment	enquiry, patho, optho, common, surgery, micro, library, radiology
9	Surgical instruments	forceps, diameter, telescope, sheath, jaw, tip, straight, bipolar

Table 4 Grouped ISO standards count

Category	ISO Standard	Count
Quality Management	ISO 9001	24
	ISO 9001:2015	16
	ISO 9001:2008	6
	ISO 9001:2000	3
Medical Devices	ISO 13,485	7
	ISO 13485:2016	3
	ISO 13485:2003	1
	ISO 10,993	2
RFID Standards	ISO 14,971	2
	ISO 14,443	10
	ISO 15,693	12
	ISO 18000:3	4
Information Security	ISO 27,001	4
	ISO 27,002	2
Miscellaneous	ISO 4628	6
	ISO 7153	3
	ISO 60	1
	ISO 4090	1
	ISO 5	4
	ISO 28,560	1
	ISO 7376:2009	1

less than 2% of all standard mentions. Notably, no occurrences were found for more domain-specific cybersecurity frameworks such as ISO/IEC 27,030 (guidelines for IoT security), ISO/IEC 30,141 (IoT reference architecture), or ISO/IEC 15,408 (Common Criteria). This pattern suggests that cybersecurity, when mentioned, is addressed primarily through generic information security management frameworks rather than device or network-specific controls. Geographical compliance references (e.g., CE: 1336 mentions; FDA: 988 mentions) further emphasize product safety and regulatory conformity as dominant concerns. Collectively, these findings indicate that while healthcare procurements are robustly aligned with quality and safety standards, cybersecurity considerations remain peripheral, appearing in roughly one in six documents where technical standards are discussed, rather than being systematically embedded in procurement specifications.

6 Gaps in cybersecurity considerations in medical devices: lessons learned and way forward

The purpose of this study was to assess the extent to which cybersecurity requirements are emphasized in the procurement of medical devices within the Indian public health-care system. Our dual corpus analysis comprises CISA Industrial Control System (ICS) advisories and Indian e-procurement documents to provide a grounded, data-driven view of the gaps between identified vulnerabilities and the institutional mechanisms designed to mitigate them.

RQ1 Are remedial actions, either technical or regulatory, addressing frequent threats, vulnerabilities, and attacker tactics, explicitly included in e-procurements of Indian medical devices?

A quantitative comparison of vulnerability themes from CISA advisories and the e-procurement corpus reveals a significant misalignment. Prevalent weaknesses in the CISA dataset, such as Improper Authentication (CWE-287) and Use of Hard-coded Credentials (CWE-798), were not matched by explicit remedial clauses in the procurement documents. Furthermore, while 61% of the analyzed vulnerabilities involved network attack vectors, in comparison, across the 760 procurement documents, only 2.6% ($n = 20$) contained references to network security or encryption, and none mentioned secure communication protocols (e.g., TLS, SSL). Similarly, there were no recorded mentions of patch or update service-level agreements, vulnerability disclosure requirements, or vendor security certifications.

This absence of specific mitigation language is further contextualized by the semantic distribution analysis (Fig. 5), where terms such as *security* and *antivirus* clustered in the “Technical-Vague” quadrant, indicating that while cybersecurity is mentioned, it is rarely operationalized through measurable requirements. Collectively, these findings suggest that security control measures are peripheral in procurement documentation, even when high-impact vulnerabilities are well established in international advisories.

RQ2 To what extent are cybersecurity requirements explicitly mentioned in e-procurement documents? Are security mechanisms addressing recent reported vulnerabilities included in these documents?

Our analysis indicates a limited explicit focus on cybersecurity. A term-frequency analysis showed that cybersecurity-related keywords appeared in only 15% of the procurement documents. When present, these terms were generic (e.g., “secure,” “authorized”) and lacked the specificity of mechanisms outlined in CISA advisories, such as those mitigating Cleartext Transmission of Sensitive Information (CWE-319). In contrast, operational terms like “warranty,” “equipment,” and “maintenance” appeared in over 80% of documents. Thematic modeling (NMF) of the procurement corpus further confirmed this operational focus, identifying prominent topics such as “Imaging and Camera Systems” and “Equipment Maintenance,” while no distinct cybersecurity theme emerged. The gathered facts, low keyword frequency, lack of specificity, and the absence of a cybersecurity topic support the conclusion that security requirements are not a primary focus and do not reflect awareness of recently reported vulnerabilities.

The Non-Negative Matrix Factorization (NMF) model reinforced this pattern. The topic coherence score for the technical specification corpus (0.7027 c_v) indicated well-defined functional clusters centered on imaging systems, diagnostics, and patient monitoring. Whereas the cybersecurity subset produced a lower coherence (0.4228 c_v), suggesting diffuse and weakly structured references to security. No dedicated cybersecurity topic emerged among the nine functional themes extracted (Table 3).

The analysis of referenced standards (Tables 4 and 5) further corroborates this pattern. Quality and safety standards such as ISO 9001 and ISO 13,485 dominate the corpus with 49 and 11 mentions, respectively, while cybersecurity-oriented standards such as ISO/IEC 27,001 and ISO/IEC 27,002 together account for fewer than 2% of all standard references. Notably, no mentions were identified for more specialized frameworks like ISO/IEC 27,030, ISO/IEC 30,141, or ISO/IEC 15,408, which directly address IoT and embedded system security. This imbalance highlights a persistent focus on product quality and regulatory conformity rather than on information or network security resilience.

Table 5 Grouped IEC standards count

Category	IEC Standard	Count
Electrical equipment	IEC 60,601	112
	IEC 601	23
	IEC 60,603	5
	IEC 950	4
Communication	IEC 11,801	12
	IEC 14,443	5
	IEC 61,034	3
Safety Standards	IEC 62,368	3
	IEC 60,529	3
	IEC 60,068	2
	IEC 60,364	2
	IEC 60,332	2
	IEC 61,010	2
Other Standards	IEC 17025:2005	1
	IEC 60,794	1
	IEC 529	1

Table 6 Example of actionable template clauses that can be embedded in e-procurement documents

Procurement document section	Proposed cybersecurity clause & evaluation criteria	Rationale & linkage to findings
Technical specifications	Clause 5.7.1 (Authentication): "The system shall not contain hard-coded credentials (CWE-798). All authentication mechanisms shall be configurable by hospital IT and support multi-factor authentication (MFA) as per NIST Level 2 (Authenticator Assurance Level 2)." Clause 5.7.2 (Data Protection): "All data transmitted to or from the device shall be encrypted using TLS 1.3. Data at rest, including stored patient data, shall be encrypted using AES-256."	Directly addresses CWE-287 and CWE-798. Replaces generic "system must be secure" with testable specifications for network vectors (61% of CISA vulnerabilities).
Vendor & contractual	Clause 17.5 (Patch SLA): "The vendor shall commit to a legally binding Service Level Agreement (SLA) defining: (a) a maximum 30-day remediation time for Critical vulnerabilities (CVSS $\geq$ 9.0), and (b) a regular security patch cycle not to exceed 90 days." Clause 17.6 (Cybersecurity Warranty): "The vendor warrants that the supplied equipment is free from known vulnerabilities at the time of delivery and will remain supported with security updates for a minimum of 10 years from procurement."	Creates an enforceable contractual mechanism for long-term security maintenance, overcoming the current lack of patch/update mandates.

Road ahead. To address the identified gaps, we propose integrating actionable, template-driven clauses into procurement frameworks that align with international standards while remaining sensitive to local implementation contexts. For example, in Table 6, integrating the "Technical Specification clauses (5.7.1, 5.7.2)" into all high-value medical device tenders (e.g., MRI, PCR systems) issued by central institutions like the AIIMS network will address the most critical assets with foundational security controls. Similarly, extending the requirements to include the "Vendor and Contractual clauses" for all new tenders will ensure cybersecurity is enforced.

Recognizing that US-focused CISA advisories may not be directly applicable in India, these measures can be localized by mapping them to ISO/IEC 27,001 and 27,002 controls, as recommended by Indian standards bodies such as the Data Security Council of India [13]. A phased adoption strategy is recommended, beginning with high-value procurements in tertiary care hospitals and gradually expanding to regional agencies,

thereby allowing organizations to build the necessary capacities for audit, compliance, and sustained implementation.

7 Conclusion

This study introduces a novel approach by leveraging procurement documents. Specifically, Indian medical procurements serve as a source of information for noting cybersecurity gaps in the healthcare sector. By comparing data from two corpora, Indian medical procurements and CISA cybersecurity advisories, we were able to create an accurate contrast between the security needs outlined by experts and those addressed in procurement processes.

Limitation of work set up. While this study provides a data-driven assessment of cybersecurity in medical device procurement, its findings should be interpreted within the scope of an exploratory text-mining analysis. The analytical methods employed include dictionary-based regex extraction, TF-IDF weighting, and Non-negative Matrix Factorization (NMF) to identify linguistic and thematic patterns rather than generate statistically validated classifications. As such, the precision of term detection depends on the breadth of the cybersecurity dictionary and the accuracy of the regex patterns used. Because procurement language varies across tenders, some implicit or context-specific cybersecurity references may not have been detected. Moreover, the study did not use a gold-standard, human-annotated corpus for validation; therefore, the results indicate thematic tendencies rather than confirmed classification accuracy.

The methodological framework reflects a deliberate trade-off between scalability and interpretive precision. Regex extraction can misclassify polysemous terms (e.g., “secure” referring to physical safety), TF-IDF can overemphasize high-frequency generic terms, and NMF topic modeling can obscure contextual nuances. Nonetheless, the combined use of these methods enabled systematic, large-scale analysis across a heterogeneous corpus. This balance was intentional, prioritizing coverage and comparability over detailed linguistic annotation in line with the exploratory aims of the study.

The scope of the dataset also imposes constraints on generalizability. The analysis was confined to publicly available Indian e-procurement data, and the derived insights may not directly extend to other healthcare ecosystems with differing regulatory or procurement frameworks. Similarly, the use of U.S.-based CISA advisories as a reference benchmark, while globally recognized, may not fully capture the threat typologies or device categories prevalent in the Indian context. Furthermore, as a document-based observational analysis, this study cannot directly infer organizational drivers, such as procurement incentives, resource limitations, or regulatory ambiguities that shape cybersecurity representation in tender documentation. The absence of standardized terminology across tenders further complicates consistent extraction and may contribute to under-detection of implicit security clauses.

Despite these limitations, the integration of regex-based term extraction, TF-IDF weighting, and NMF topic modeling strengthens the interpretive validity of the results. This multi-method approach enables the analysis to go beyond surface-level keyword counts and to reveal structural and semantic gaps in how cybersecurity is represented in procurement language. The proposed work is transparent, scalable, and reproducible, offering a methodological baseline for future computational audits of cybersecurity integration in public procurement.

Future work. Future research could extend this work by incorporating cross-country corpora, developing domain-specific gold-standard datasets, and engaging with procurement stakeholders to triangulate textual findings with institutional practice. Longitudinal analyses and LLM-assisted semantic enrichment could further enable the continuous monitoring of cybersecurity maturity in procurement documentation. Such efforts would support the development of standardized, security-aware procurement frameworks, advancing resilience in the medical device ecosystem within the broader Internet of Things (IoT) landscape.

Author contributions

Dr Kumar contributed with the study conception, resource acquisition and design. Material preparation, data collection and analysis were jointly performed by R. Kumar, and S. Pandey. The first draft of the manuscript was written by S. Pandey and was subsequently revised by Dr Kumar. All authors read and approved the final manuscript.

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Data availability

All data and code is available at: <https://github.com/mrcodeScholar/Cybersecurity-of-Health-care-devices> (<https://github.com/mrcodeScholar/Cybersecurity-of-Health-care-devices>).

Declarations

Competing interests

The authors declare no competing interests.

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